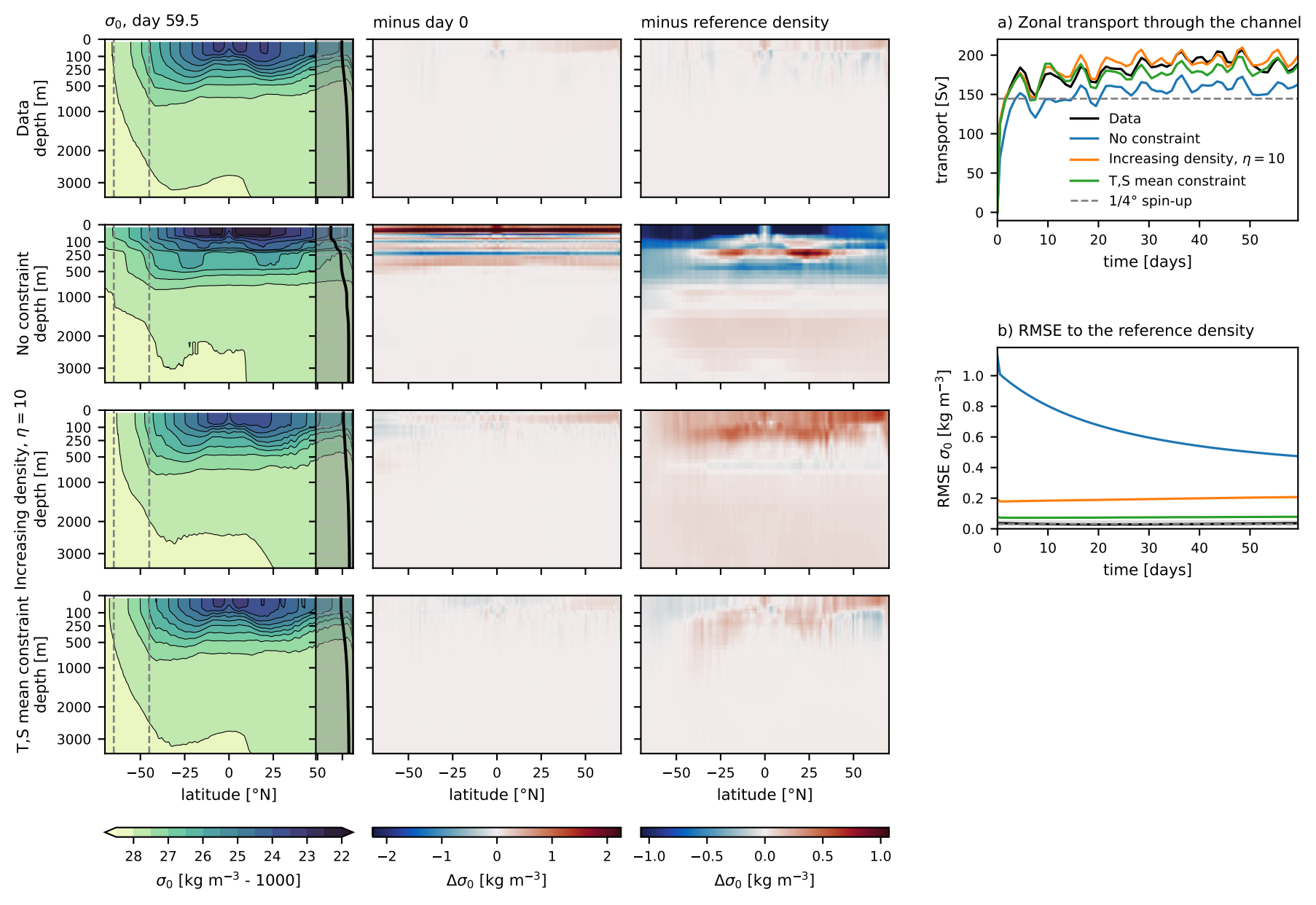

σ_0, day 59.5
minus day 0
minus reference density
Data
No constraint
Increasing density, $\eta = 10$
T,S mean constraint
depth [m]
0
100
250
500
1000
2000
3000
−50
−25
0
25
50
latitude [°N]
28
27
26
25
24
23
22
σ_0 [kg m^{-3} - 1000]
−2
−1
0
1
2
$\Delta\sigma_0$ [kg m^{-3}]
−1.0
−0.5
0.0
0.5
1.0
$\Delta\sigma_0$ [kg m^{-3}]
a) Zonal transport through the channel
transport [Sv]
200
150
100
50
0
Data
No constraint
Increasing density, $\eta = 10$
T,S mean constraint
1/4° spin-up
0
10
20
30
40
50
time [days]
b) RMSE to the reference density
RMSE σ_0 [kg m^{-3}]
1.0
0.8
0.6
0.4
0.2
0.0
time [days]